

Table of Contents

Task 1 — Reconnaissance	3
Recon-ng	3
Workspace Management	3
Module Marketplace	3
Module Execution (Information Gathering)	4
Nmap	5
Host Discovery (Ping Scan)	5
Service and Version Detection	5
OS Fingerprinting	5
Hping3	6
TCP SYN Scan	6
ICMP Ping Test	6
UDP Scan	6
DNSRecon	7
DNS Record Enumeration	7
Zone Transfer Attempt	8
Reverse Lookup	8
Task 2 — Maintaining Access	9
PowerSploit	9
Module Loading	9
Credential Access	9
Privilege Escalation Module Access	9
Why PowerSploit Failed in Kali Linux	9
Webshells	10
Remote Command Execution	10
System Information Retrieval	10
File System Interaction	10
Weevely	11
Backdoor Agent Generation	11
Remote Terminal Connection	11
Internal System Enumeration	11

Dns2tcp	12
Configuration File Setup.....	12
Server Setup (dns2tcpd).....	12
Client Concept	13
Cryptcat.....	13
Encrypted Listener Setup (Server Side).....	13
Remote Connection (Client Side)	13
Encrypted Data Transfer	13
Conclusion	14

Task 1 — Reconnaissance

Reconnaissance tools are used to collect information about a target before attempting any form of exploitation.

Recon-ng

Recon-ng is a reconnaissance framework used to gather information about a target through automated modules, helping identify domains, hosts, and other related data.

Workspace Management

```
(kali@kali)-[~]
$ recon-ng
[*] Version check disabled.
```



Sponsored by ...



```
[recon-ng v5.1.2, Tim Tomes (@lanmaster53)]

[*] No modules enabled/installed.

[recon-ng][default] > workspaces create test_lab
[recon-ng][test_lab] > workspaces list
```

Workspaces	Modified
default	2026-02-27 10:02:12
test	2026-02-27 10:03:58
test_lab	2026-05-06 10:13:27

This feature is used to create and manage different workspaces for each reconnaissance project.

Module Marketplace

```
[recon-ng][test_lab] > marketplace install recon/domains-hosts/google_site_web
[*] Module installed: recon/domains-hosts/google_site_web
[*] Reloading modules ...
```

This command allows users to install and manage modules within Recon-ng.

Module Execution (Information Gathering)

```
[recon-ng][test_lab] > marketplace install recon/domains-hosts/google_site_web
[*] Module installed: recon/domains-hosts/google_site_web
[*] Reloading modules ...
[recon-ng][test_lab] > modules load recon/domains-hosts/google_site_web
[recon-ng][test_lab][google_site_web] > options set SOURCE example.com
SOURCE ⇒ example.com
[recon-ng][test_lab][google_site_web] > run

_____
EXAMPLE.COM
_____
[*] Searching Google for: site:example.com
```

This feature is used to load and run specific modules to collect data about a target domain.

Nmap

Nmap is a widely-used network scanning tool for discovering hosts, ports and services on a network.

Host Discovery (Ping Scan)

```
(kali㉿kali)-[~]  
$ nmap -sn 127.0.0.1  
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-27 09:35 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.00012s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

This feature identifies which devices are currently online without performing a heavy port scan, allowing for a faster and stealthier initial map of the target network.

Service and Version Detection

```
(kali㉿kali)-[~]  
$ nmap -sV 127.0.0.1  
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-27 10:00 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000092s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (conn-refused)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.81 seconds
```

This scan gathers the version information for each open service to determine if the service is outdated or contains known vulnerabilities.

OS Fingerprinting

```
(kali㉿kali)-[~]  
$ sudo nmap -O 127.0.0.1  
Starting Nmap 7.92 ( https://nmap.org ) at 2026-02-27 10:28 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000057s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.09 seconds
```

This command is used to identify the operating system (like Windows, Linux or macOS) of a target device by analysing its unique network behaviour.

Hping3

Hping3 is a network tool used to create and send custom packets for testing firewall rules, network performance, and security.

TCP SYN Scan

```
(kali㉿kali)-[~]
└─$ sudo hping3 -S example.com -p 80
[sudo] password for kali:
HPING example.com (eth0 172.66.147.243): S set, 40 headers + 0 data bytes
len=46 ip=172.66.147.243 ttl=64 id=36804 sport=80 flags=SA seq=0 win=65535 rtt=27.5 ms
len=46 ip=172.66.147.243 ttl=64 id=43190 sport=80 flags=SA seq=1 win=65535 rtt=26.9 ms
len=46 ip=172.66.147.243 ttl=64 id=21546 sport=80 flags=SA seq=2 win=65535 rtt=31.4 ms
len=46 ip=172.66.147.243 ttl=64 id=42287 sport=80 flags=SA seq=3 win=65535 rtt=30.5 ms
len=46 ip=172.66.147.243 ttl=64 id=47651 sport=80 flags=SA seq=4 win=65535 rtt=34.1 ms
len=46 ip=172.66.147.243 ttl=64 id=26995 sport=80 flags=SA seq=5 win=65535 rtt=29.6 ms
len=46 ip=172.66.147.243 ttl=64 id=6406 sport=80 flags=SA seq=6 win=65535 rtt=24.4 ms
len=46 ip=172.66.147.243 ttl=64 id=958 sport=80 flags=SA seq=7 win=65535 rtt=24.4 ms
len=46 ip=172.66.147.243 ttl=64 id=53264 sport=80 flags=SA seq=8 win=65535 rtt=35.5 ms
^C
--- example.com hping statistic ---
9 packets transmitted, 9 packets received, 0% packet loss
round-trip min/avg/max = 24.4/29.4/35.5 ms
```

This command is used to send TCP SYN packets to a target port to check if the port is open.

ICMP Ping Test

```
(kali㉿kali)-[~]
└─$ sudo hping3 -I example.com
HPING example.com (eth0 104.20.23.154): icmp mode set, 28 headers + 0 data bytes
len=46 ip=104.20.23.154 ttl=255 id=65231 icmp_seq=0 rtt=27.9 ms
len=46 ip=104.20.23.154 ttl=255 id=890 icmp_seq=1 rtt=28.2 ms
len=46 ip=104.20.23.154 ttl=255 id=24304 icmp_seq=2 rtt=32.3 ms
len=46 ip=104.20.23.154 ttl=255 id=7799 icmp_seq=3 rtt=25.7 ms
len=46 ip=104.20.23.154 ttl=255 id=12864 icmp_seq=4 rtt=29.1 ms
len=46 ip=104.20.23.154 ttl=255 id=2381 icmp_seq=5 rtt=29.2 ms
len=46 ip=104.20.23.154 ttl=255 id=42056 icmp_seq=6 rtt=32.0 ms
^C
--- example.com hping statistic ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 25.7/29.2/32.3 ms
```

This feature sends ICMP packets to check if a host is reachable.

UDP Scan

```
(kali㉿kali)-[~]
└─$ sudo hping3 -2 example.com -p 53
HPING example.com (eth0 172.66.147.243): udp mode set, 28 headers + 0 data bytes
^C
--- example.com hping statistic ---
24 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

This feature sends UDP packets to a specific port to determine if the service is available. The result shows 100% packet loss because UDP does not always return a response.

DNSRecon

DNSRecon is a tool used to gather DNS information about a domain, such as records, subdomains and name servers.

DNS Record Enumeration

```
(kali@kali)-[~]
$ dnsrecon -d example.com
[*] std: Performing General Enumeration against: example.com...
[*] DNSSEC is configured for example.com
[*] DNSKEYs:
[*] NSEC ZSK ECDSA256SHA256 9318a98e821b3590ec5aa10a698686ab a7ccfd74e1ad1a75b9ee8057d47f9f77 96c691827825896f8ecf104123fd0e95 018156f
fb505a39997a197f129865adc
[*] NSEC KSK ECDSA256SHA256 99db2cc14cabdc33d6d77da63a2f15f7 1112584f234e8d1dc428e39e8a4a97e1 aa271a555dc90701e17e2a4c4b6f120b 7c32d44
f4ac02bd894cf2d4be7778a19
[*] NSEC ZSK ECDSA256SHA256 323c9989e94fd06aa7888d7e8fec001b fdedd089830c8963d42c51d28a1b9194 0748a1f616ac7ab66d8d718bb9ecf389 f374b30
962e4a888d91d2d66879c1b6b
[*] NSEC ZSK ECDSA256SHA256 a09311112cf9138818cd2feae970ebbd 4d6a30f6088c25b325a39abbc5cd1197 aa098283e5aaf421177c2aa5d714992a 9957d1b
cc18f98cd71f1f1806b65e148
[*] SOA elliottns.cloudflare.com 172.64.35.228
[*] SOA elliottns.cloudflare.com 162.159.44.228
[*] SOA elliottns.cloudflare.com 108.162.195.228
[*] SOA elliottns.cloudflare.com 2803:f800:50::6ca2:c3e4
[*] SOA elliottns.cloudflare.com 2606:4700:58::a29f:2ce4
[*] SOA elliottns.cloudflare.com 2a06:98c1:50::ac40:23e4
[*] NS herans.cloudflare.com 173.245.58.162
[*] NS herans.cloudflare.com 108.162.192.162
[*] NS herans.cloudflare.com 172.64.32.162
[*] NS herans.cloudflare.com 2a06:98c1:50::ac40:20a2
[*] NS herans.cloudflare.com 2606:4700:50::adf5:3aa2
[*] NS herans.cloudflare.com 2803:f800:50::6ca2:c0a2
[*] NS elliottns.cloudflare.com 108.162.195.228
[*] NS elliottns.cloudflare.com 162.159.44.228
[*] NS elliottns.cloudflare.com 172.64.35.228
[*] NS elliottns.cloudflare.com 2a06:98c1:50::ac40:23e4
[*] NS elliottns.cloudflare.com 2606:4700:58::a29f:2ce4
[*] NS elliottns.cloudflare.com 2803:f800:50::6ca2:c3e4
[*] A example.com 172.66.147.243
[*] A example.com 104.20.23.154
[*] AAAA example.com 2606:4700:10::ac42:93f3
[*] AAAA example.com 2606:4700:10::6814:179a
[*] TXT example.com v=spf1 -all
[*] TXT example.com _k2n1y4vw3qtb4skdx9e7dxt97qrmmq9
[*] TXT _dmarc.example.com v=DMARC1;p=reject;sp=reject;adkim=s;aspf=s
[*] TXT _dmarc._domainkey.example.com v=DKIM1; p=
[*] Enumerating SRV Records
[*] 0 Records Found
```

This feature retrieves standard DNS records such as A, MX, and NS records. It helps identify important information about the target domain's infrastructure.

Zone Transfer Attempt

```
(kali㉿kali)-[~]
└─$ dnsrecon -d example.com -t axfr
[*] Checking for Zone Transfer for example.com name servers
[*] Resolving SOA Record
[+] SOA elliottns.cloudflare.com 108.162.195.228
[+] SOA elliottns.cloudflare.com 162.159.44.228
[+] SOA elliottns.cloudflare.com 172.64.35.228
[+] SOA elliottns.cloudflare.com 2a06:98c1:50::ac40:23e4
[+] SOA elliottns.cloudflare.com 2803:f800:50::6ca2:c3e4
[+] SOA elliottns.cloudflare.com 2606:4700:58::a29f:2ce4
[*] Resolving NS Records
[*] NS Servers found:
[+] NS elliottns.cloudflare.com 172.64.35.228
[+] NS elliottns.cloudflare.com 162.159.44.228
[+] NS elliottns.cloudflare.com 108.162.195.228
[+] NS elliottns.cloudflare.com 2606:4700:58::a29f:2ce4
[+] NS elliottns.cloudflare.com 2803:f800:50::6ca2:c3e4
[+] NS elliottns.cloudflare.com 2a06:98c1:50::ac40:23e4
[+] NS herans.cloudflare.com 172.64.32.162
[+] NS herans.cloudflare.com 108.162.192.162
[+] NS herans.cloudflare.com 173.245.58.162
[+] NS herans.cloudflare.com 2606:4700:50::adf5:3aa2
[+] NS herans.cloudflare.com 2803:f800:50::6ca2:c0a2
[+] NS herans.cloudflare.com 2a06:98c1:50::ac40:20a2
[*] Removing any duplicate NS server IP Addresses ...
[*]
[*] Trying NS server 173.245.58.162
[+] 173.245.58.162 Has port 53 TCP Open
[-] Zone Transfer Failed (Zone transfer error: FORMERR)
```

This feature attempts to perform a DNS zone transfer from the target name server.

Reverse Lookup

```
(kali㉿kali)-[~]
└─$ dnsrecon -r 127.0.0.1/24
[*] Performing Reverse Lookup from 127.0.0.0 to 127.0.0.255
[+] 0 Records Found

(kali㉿kali)-[~]
└─$ dnsrecon -r localhost
[-] Range provided is not valid
[-] Invalid Address/CIDR or Address Range provided.
```

This feature performs a reverse DNS lookup to find domain names linked to IP addresses.

Task 2 — Maintaining Access

The Maintaining Access phase is critical for ensuring that a penetration tester can remain inside a target system for an extended period, even if the system is rebooted or the initial vulnerability is patched.

PowerSploit

PowerSploit is a Windows-based PowerShell post-exploitation framework used for privilege escalation, credential extraction, and maintaining access after system compromise.

Module Loading

```
(kali@kali)-[/usr/share/windows-resources/powersploit]
PS> Import-Module ./PowerSploit.ps1
InvalidOperation: /usr/share/windows-resources/powersploit/Recon/PowerView.ps1:68
Line
68      $AssemblyBuilder = $Domain.DefineDynamicAssembly($DynAssembly, 'R ...
      ~~~~~
      Method invocation failed because [System.AppDomain] does not contain a method named 'DefineDynamicAssembly'.
InvalidOperation: /usr/share/windows-resources/powersploit/Recon/PowerView.ps1:69
Line
69      $ModuleBuilder = $AssemblyBuilder.DefineDynamicModule($ModuleName ...
      ~~~~~
      You cannot call a method on a null-valued expression.
```

This feature is used to load the PowerSploit framework into PowerShell so that its modules can be accessed.

Credential Access

```
(kali@kali)-[/usr/share/windows-resources/powersploit]
PS> Import-Module ./Recon/PowerView.ps1
InvalidOperation: /usr/share/windows-resources/powersploit/Recon/PowerView.ps1:68
Line
68      $AssemblyBuilder = $Domain.DefineDynamicAssembly($DynAssembly, 'R ...
      ~~~~~
      Method invocation failed because [System.AppDomain] does not contain a method named 'DefineDynamicAssembly'.
InvalidOperation: /usr/share/windows-resources/powersploit/Recon/PowerView.ps1:69
Line
69      $ModuleBuilder = $AssemblyBuilder.DefineDynamicModule($ModuleName ...
      ~~~~~
      You cannot call a method on a null-valued expression.
```

This command is used to perform reconnaissance on a Windows domain environment.

Privilege Escalation Module Access

```
(kali@kali)-[/usr/share/windows-resources/powersploit]
PS> Import-Module ./Privesc/PowerUp.ps1
```

This feature provides tools to identify privilege escalation opportunities in a system.

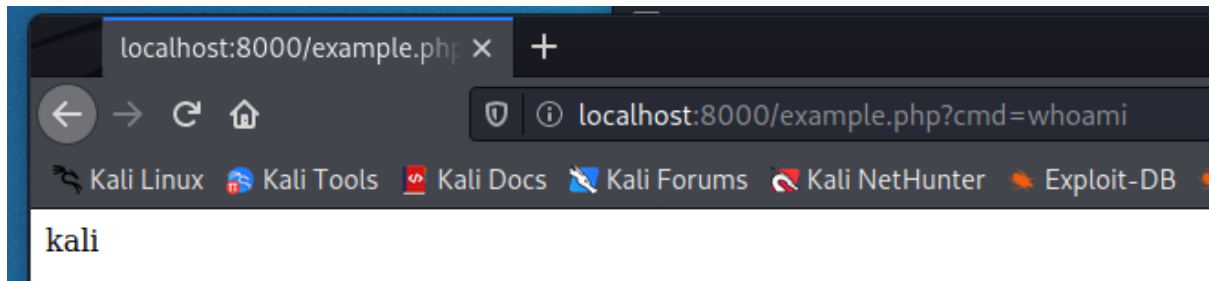
Why PowerSploit Failed in Kali Linux

PowerSploit cannot fully run in Kali Linux because it depends on Windows-specific .NET and PowerShell features that are not fully supported in Linux-based PowerShell environments.

Webshells

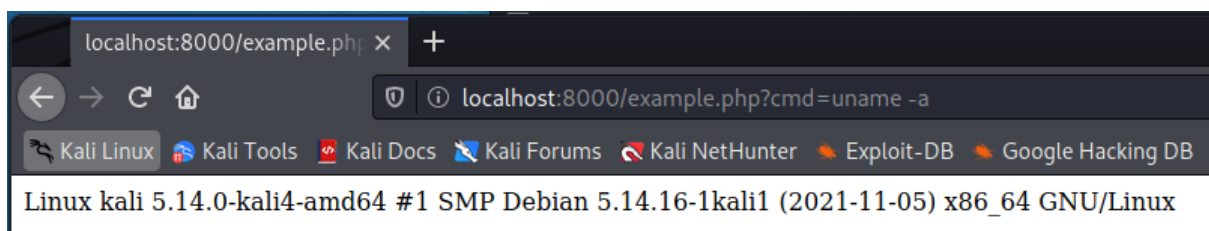
Webshells are malicious scripts uploaded to a web server that allow remote command execution and control over the target system through a web interface.

Remote Command Execution



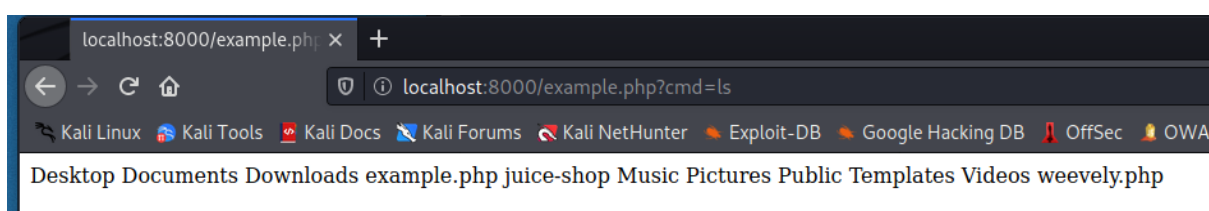
This feature allows execution of system commands through a web browser using URL parameters.

System Information Retrieval



This feature is used to retrieve system information such as operating system details and kernel version through browser-based commands.

File System Interaction



This feature allows browsing and listing files in the server directory through web-based command execution.

Weevely

Weevely is a stealthy PHP web shell used to maintain a persistent, telnet-like connection to a compromised web server.

Backdoor Agent Generation

```
(kali㉿kali)-[~]  
$ weevely generate secret weevely.php  
Generated 'weevely.php' with password 'secret' of 697 byte size.
```

This creates a small, obfuscated PHP script that acts as the "agent." Its polymorphic nature makes it difficult for antivirus software to detect.

Remote Terminal Connection

```
(kali㉿kali)-[~]  
$ weevely http://localhost/weevely.php secret  
  
[+] weevely 4.0.1  
[+] Target:      localhost  
[+] Session:     .weevely/sessions/localhost/weevely_0.session  
  
[+] Browse the filesystem or execute commands starts the connection  
[+] to the target. Type :help for more information.
```

This feature establishes a persistent communication channel over HTTP. By tunnelling the shell access through web traffic, it allows the tester to maintain access through firewalls that only allow web traffic (Port 80/443), making the remote-control session appear as legitimate web activity.

Internal System Enumeration

```
weevely> :system_info  
+-----+  
| document_root | /var/www/html |  
| whoami        | www-data      |  
| hostname      |               |  
| pwd           | /var/www/html |  
| open_basedir  |               |  
| safe_mode     | False         |  
| script        | /weevely.php  |  
| script_folder | /var/www/html |  
| uname         | Linux kali 5.14.0-kali4-amd64 #1 SMP Debian 5.14.16-1kali1 (2021-11-05) x86_64 |  
| os            | Linux         |  
| client_ip     | ::1           |  
| max_execution_time | 30          |  
| php_self      | /weevely.php  |  
| dir_sep       | /             |  
| php_version   | 7.4.21        |  
+-----+
```

This is used to profile the target environment by identifying server configurations, PHP restrictions and operating system details necessary for further administration or post-exploitation.

Dns2tcp

Dns2tcp is a tunneling tool that allows data and connections to be transferred through DNS protocol, often used to bypass firewalls and maintain covert communication.

Configuration File Setup

```
(kali㉿kali)-[~]  
$ cat dns2tcp.conf  
listen = 127.0.0.1  
port = 53  
user = test  
key = secretkey  
domain = test.local  
resources = ssh:127.0.0.1:22  
  
(kali㉿kali)-[~]  
$ dns2tcpd -f dns2tcp.conf
```

This configuration defines the DNS tunneling environment including authentication, domain name, and available services such as SSH forwarding.

Server Setup (dns2tcpd)

```
(kali㉿kali)-[~]  
$ dns2tcpd -F -f dns2tcp.conf  
Starting Server v0.5.2 ...  
User unknown test
```

This command starts the dns2tcp server in foreground mode using a configuration file.

Client Concept

```
(kali㉿kali)-[~]  
$ dns2tcp -r ssh -z test.local 127.0.0.1
```

This feature represents the client-side connection to the dns2tcp server. It uses DNS queries to establish a tunnel and access services such as SSH through restricted networks.

Cryptcat

Cryptcat is a secure version of Netcat that provides encrypted communication between a client and a server, often used for remote shell access and data transfer.

Encrypted Listener Setup (Server Side)

```
(kali㉿kali)-[~]  
$ cryptcat -l -p 4444 -k secret
```

This feature starts a listening port on the target system. It waits for incoming encrypted connections from a client using a shared password key.

Remote Connection (Client Side)

```
(kali㉿kali)-[~]  
$ cryptcat 127.0.0.1 4444 -k secret
```

This feature connects to the listening server using an encrypted channel.

Encrypted Data Transfer

```
(kali㉿kali)-[~]  
$ echo "hello" | cryptcat 127.0.0.1 4444 -k secret
```

The command `echo "hello" | cryptcat` demonstrates encrypted data transfer by piping plain text into Cryptcat, which then encrypts the data using a shared key before sending it over the network connection.

Conclusion

Tool	Purpose	Type	Key Strength
Recon-ng	Information gathering	Reconnaissance	Automated OSINT modules
Nmap	Network scanning	Reconnaissance	Fast port and service detection
DNSRecon	DNS enumeration	Reconnaissance	Subdomain and DNS discovery
Hping3	Packet crafting	Network testing	Custom packet control
PowerSploit	Post-exploitation	Windows framework	Privilege escalation tools
Webshells	Remote command execution	Post-exploitation	Browser-based control
Weevely	Web shell management	Post-exploitation	Lightweight PHP backdoor control
dns2tcp	DNS tunneling	Post-exploitation	Firewall bypass via DNS
Cryptcat	Secure communication	Post-exploitation	Encrypted data transfer

In conclusion, the tools used in this assignment cover both reconnaissance and maintaining access stages of penetration testing. Reconnaissance tools such as Recon-ng, Nmap, and DNSRecon are used to collect information about target systems, while tools like Webshells, Cryptcat, dns2tcp, and Weevely focus on maintaining access and enabling remote control. Throughout the practical work, some tools such as PowerSploit and dns2tcp showed compatibility and configuration challenges in the Kali Linux environment. This highlights that effective penetration testing requires not only tool usage, but also understanding of tool behaviour, troubleshooting skills and system compatibility.